HEALTHCARE PATIENT MANAGEMENT SYSTEM

Enterprise Microservice & Distributed Relational Architecture

DATA INGESTION (ETL)

Raw & Ingestion Layer

- UCI 130-US Hospitals
- 100K+ Inpatient Records
- Pandas Cleaning Engine
- Deduplication Filter
- Missing Value Imputer
- ICD-9 Normalizer
- Incremental Loader (CDC)

STORAGE LAYER

Relational Core

- Dual-Engine Connector
- MySQL (Production)
- SQLite (Zero-Friction)
- 3NF Relational Model
- patients (Master)
- encounters (Fact)
- encounter_diagnoses
- encounter_medications
- audit_logs (Compliance)

MICROSERVICE (API)

FastAPI Service Layer

- OAuth2 + JWT Auth
- Argon2 Security
- Role-Based Access (RBAC)
- Admin / Staff / Analyst
- Patient & Encounter CRUD
- Aggregation Engine
- CSV / Excel Exporters
- Audit Interceptor

PRESENTATION LAYER

Streamlit Clinical UI

- Deep Navy & Teal Theme
- Role-Gated Navigation
- KPI Performance Cards
- Readmission Analytics
- Length of Stay Histograms
- Patient Timeline Viewer
- Admission Operations
- Audit Compliance Log

ENTERPRISE SECURITY & AUDIT TRAIL ARCHITECTURE

Strict Role Enforcement: Admin (User Admin, Audits) | Healthcare Staff (Patient CRUD, Admissions) | Analyst (Aggregations, CSV/Excel Reports)